

Improved UNet3D for Prostate MRI Segmentation with Attention, Residual, and ASPP Modules

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This repository presents the development, data augmentation strategies, and training pipeline for an Improved UNet3D architecture, designed for 3D medical image segmentation on the Prostate 3D dataset. The model builds upon the foundational U-Net architecture which has become one of the most influential convolutional network designs in biomedical image segmentation. The improved version implemented here integrates advanced components such as residual connections, squeeze-and-excitation (SE) blocks, attention gates, and atrous spatial pyramid pooling (ASPP) to enhance feature representation, gradient flow, and multi-scale context capture in volumetric data.

For detailed insights into the original 3D U-Net design and medical imaging applications, please refer to the paper: [Few-Shot Learning for Medical Image Segmentation Using 3D U-Net and Model-Agnostic Meta-Learning \(MAML\)](#)

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Data Set

The project uses the downsampled Prostate 3D dataset consisting of paired MRI volumes and segmentation masks stored in NIfTI format ([.nii.gz](#)).

Each image–label pair is identified using matching filename keys (`*_LFOV` for MRI, `*_SEMANTIC` for label). When loaded via the `HipMRI3DDataset` class:

- Volumes are z-score normalized (per subject).
- Patches are center-cropped or padded to a fixed shape (default: 256×256×128).
- Labels are converted to `torch.LongTensor` and images to `torch.FloatTensor`. Data pairs are automatically matched using the `find_pairs()` utility, which scans the image and label directories and returns valid `(key, image_path, label_path)` tuples.

Project Goal

The goal of this project is to segment the downsampled Prostate 3D MRI dataset using an **Improved UNet3D** model that integrates **residual connections**, **squeeze-and-excitation (SE) blocks**, **attention gates**, and **ASPP modules**.

The model aims to achieve a minimum Dice Similarity Coefficient (DSC) of **0.7** for all labels on the test set, ensuring accurate and reliable prostate boundary segmentation.

File Structure

This repository consists of the following four major files:

- `dataset.py` - Loads and preprocesses images, returns tensors and labels for training and validation.
- `modules.py` - Defines model architectures, losses, and metric functions used during training and evaluation.
- `train.py` - Training and validation loop, handles optimizer/scheduler, checkpointing, and logging.
- `predict.py` - Loads a trained checkpoint and runs inference on new inputs

Model Architecture

Base UNet3D Overview

The **Improved UNet3D** follows the encoder–decoder design of the classic U-Net, adapted for 3D medical imaging.

The encoder captures hierarchical features through downsampling, while the decoder restores spatial resolution via upsampling and skip connections.

Channel depth starts at **32** and doubles at each level (32 → 64 → 128 → 256 → 512).

An **ASPP** block at the bottleneck aggregates multi-scale context before decoding.

Residual Connections

Each stage uses **residual blocks** (`Res3D`) with two 3D convolutions, **Group Normalization**, and **SiLU** activation.

Residual shortcuts improve gradient flow, training stability, and representation depth, allowing the network to learn finer prostate details.

Squeeze-and-Excitation (SE) Blocks

SE3D modules adaptively recalibrate channel responses.

Global average pooling captures channel statistics, which are used to reweight features via learned attention.

This focuses the network on the most informative tissue structures and suppresses noise.

Attention Gates

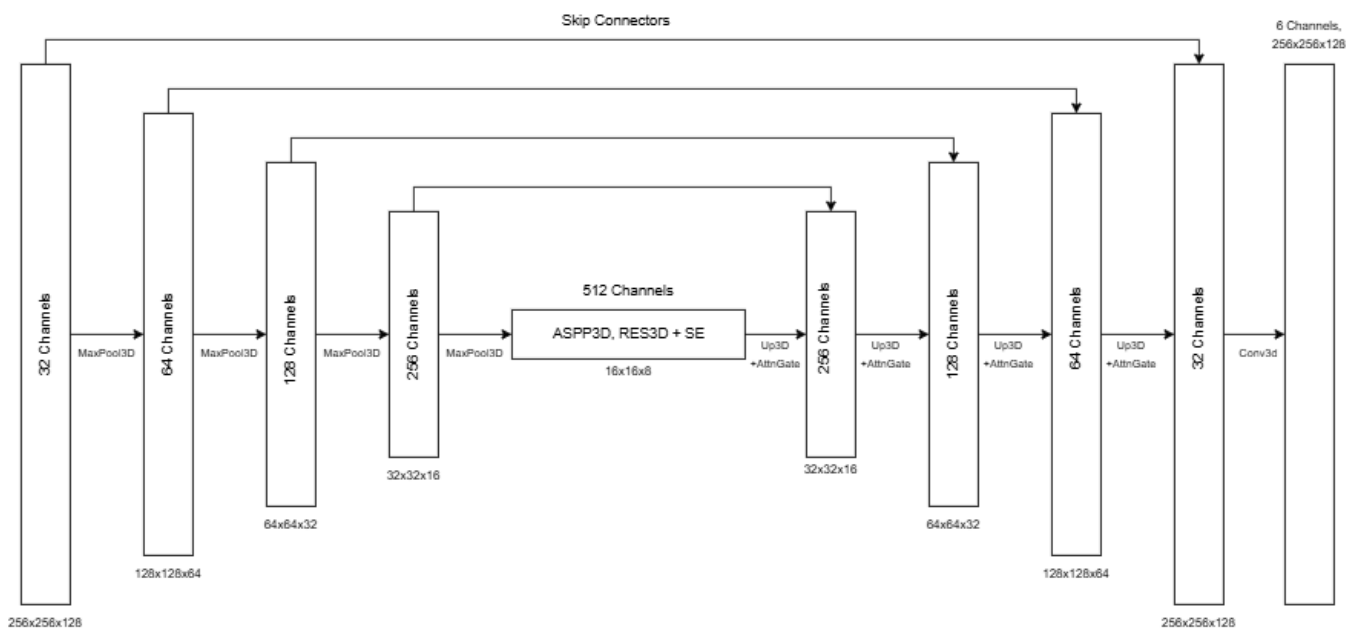
Decoder **attention gates** refine skip connections by weighting spatial regions based on relevance. Each gate combines encoder and decoder features to generate an attention mask, filtering out irrelevant background and highlighting key anatomical areas for fusion.

Atrous Spatial Pyramid Pooling (ASPP)

The **ASPP3D** module captures multi-scale context using parallel dilated convolutions with rates (1, 2, 4, 6) and a global pooling branch.

These are concatenated and projected into a unified feature map, enhancing robustness to prostate size and shape variation.

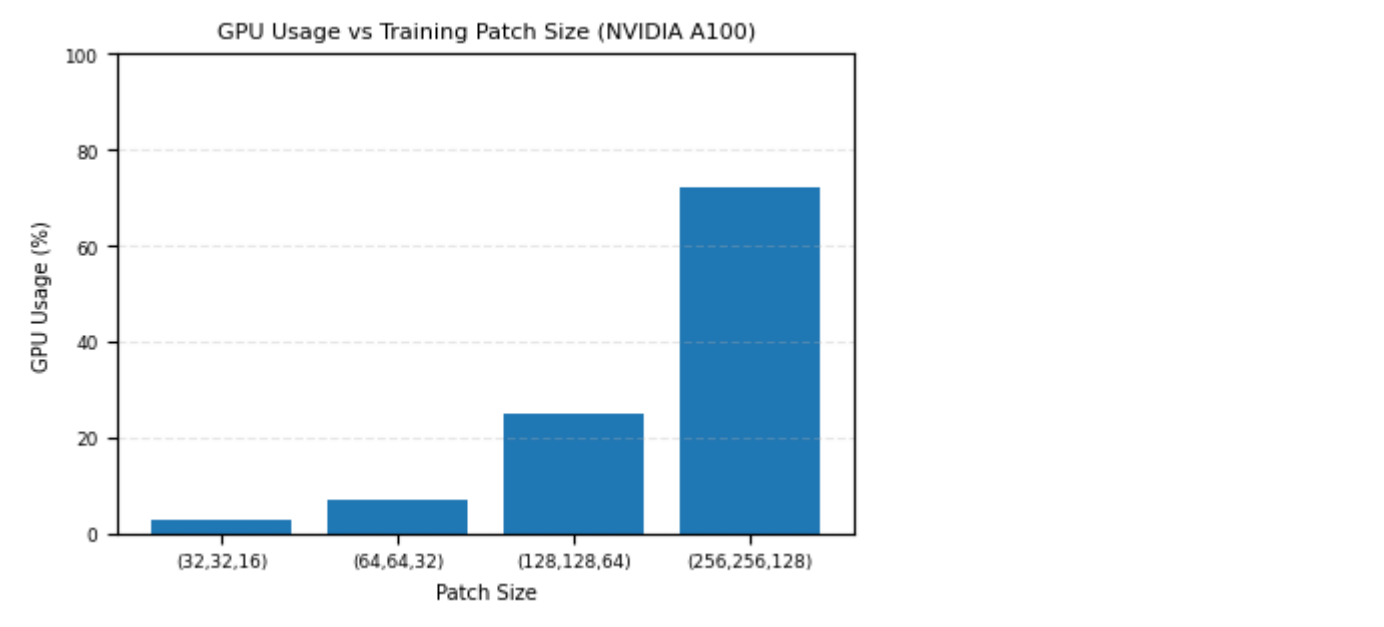
Model Diagram



Training

Initially, I tested **different patch sizes** while keeping `batch_size=1` and `num_workers=4` to observe GPU utilization and determine the optimal configuration for my A100 cluster. As illustrated below, GPU usage increased almost linearly with the size of the training patch.

After testing, I settled on a patch size of **(256, 256, 128)**, which reached around **72% GPU utilization** on the **NVIDIA A100** without exceeding memory limits. I then increased the number of data-loading workers to **8** (the maximum available cores) to minimize data-loading bottlenecks during training.



Training Configuration

Training was performed using the following parameters:

Parameter	Value	Description
PATCH_SIZE	(256, 256, 128)	Target volume size for training
BATCH_SIZE	1	Single-volume per batch
ACCUM_STEPS	4	Gradient accumulation to simulate batch size of 4
BASE_CH	32	Base convolution channel size
DROPOUT	0.1	Dropout regularization
NUM_EPOCHS	30	Total training epochs
LR	1e-4	Learning rate
WEIGHT_DECAY	1e-5	AdamW regularization
NUM_WORKERS	8	Parallel data-loading workers
AMP	Enabled	Automatic mixed precision for faster GPU training
SAVE_PATH	UNET3D_HIPMRI_BEST.PT	Checkpoint for best validation model

The **Improved UNet3D** integrates residual connections, squeeze-and-excitation (SE) blocks, attention gates, and ASPP modules.

The loss function combines **CrossEntropyLoss** and **DiceLoss3D** equally weighted to balance pixel-level and shape-level supervision:

- $$\mathcal{L} = 0.5 \cdot \text{CrossEntropy} + 0.5 \cdot \text{DiceLoss3D}$$

Gradient accumulation and **mixed precision (AMP)** allowed training large 3D patches efficiently within GPU memory limits.

Validation and Testing

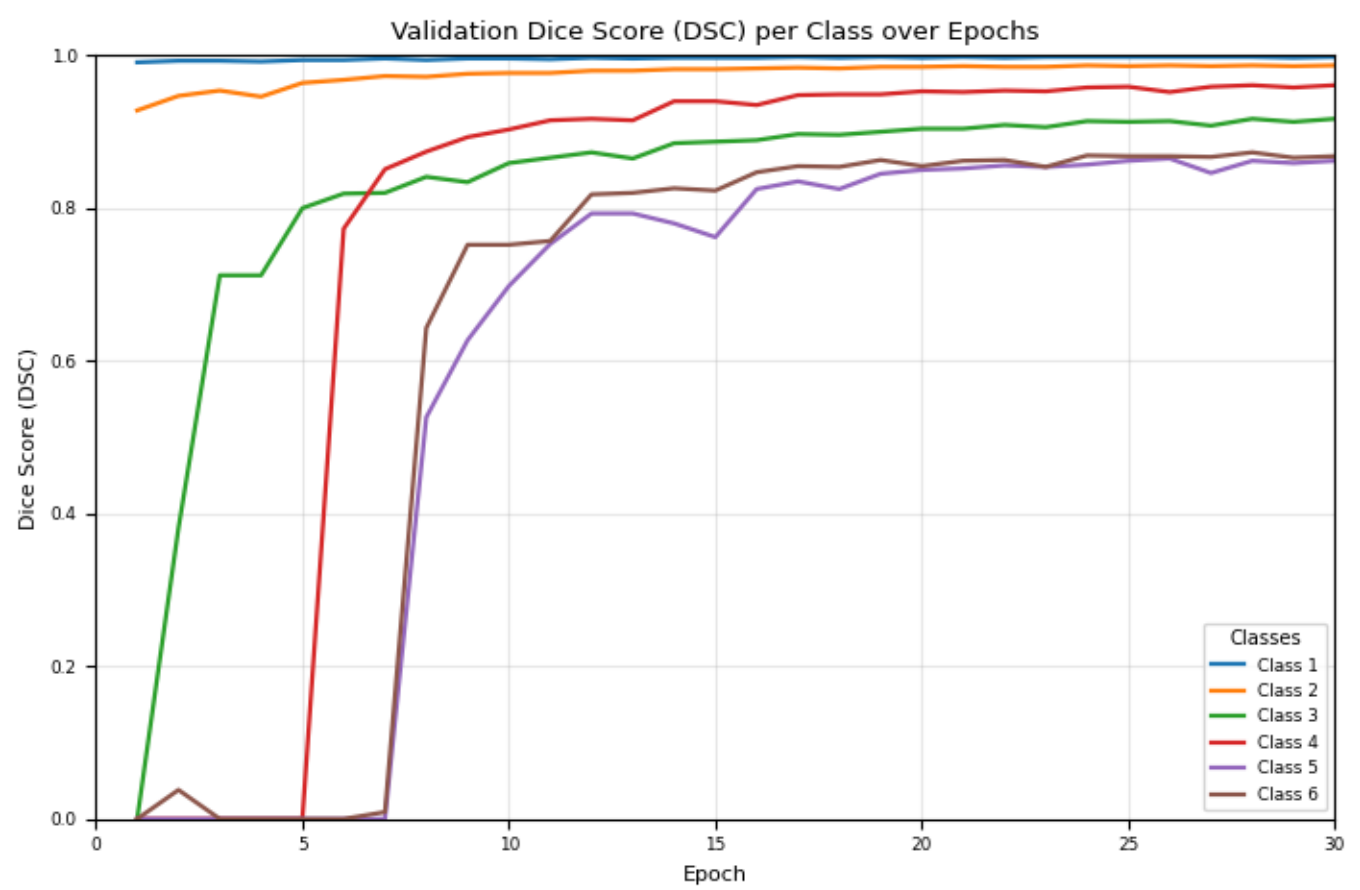
Each epoch reported training loss, validation loss, and **Dice Similarity Coefficient (DSC)** per class. The best checkpoint was saved automatically when validation loss improved.

Final training and validation results after 30 epochs:

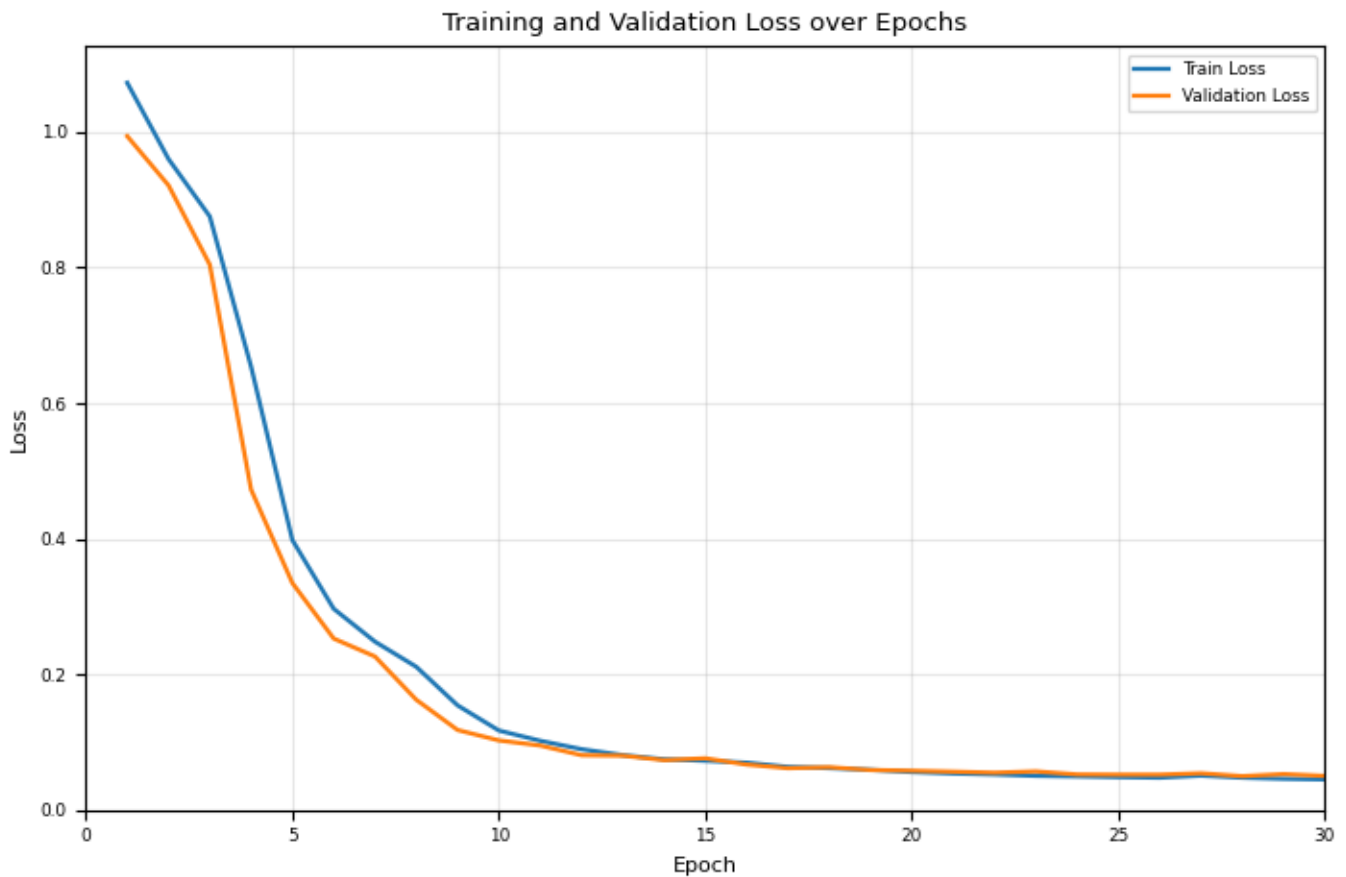
```
Train Loss: 0.0450 | Val Loss: 0.0501
Val DSC: [0.998, 0.987, 0.917, 0.961, 0.862, 0.868]
```

Validation Performance Over Epochs

The figure below shows the **Dice Score (DSC)** evolution across 30 epochs for each label. Major structures reached stability early ($\approx$ epoch 10), while finer structures converged gradually between epochs 15–25.



The figure below shows that both training and validation losses decrease rapidly during the initial epochs and gradually stabilize after around 10 epochs. This close convergence indicates that the model has learned effectively with minimal overfitting and strong generalization performance.



- Increasing the patch size improved spatial context capture and prostate boundary precision.
- Setting `num_workers=8` significantly reduced data-loading latency and improved epoch runtime.
- The Improved UNet3D achieved **high segmentation accuracy**, surpassing the project target of **DSC ≥ 0.7** for all labels.
- Smaller structures benefited most from the **attention and ASPP** modules, leading to smoother convergence and improved boundary delineation.

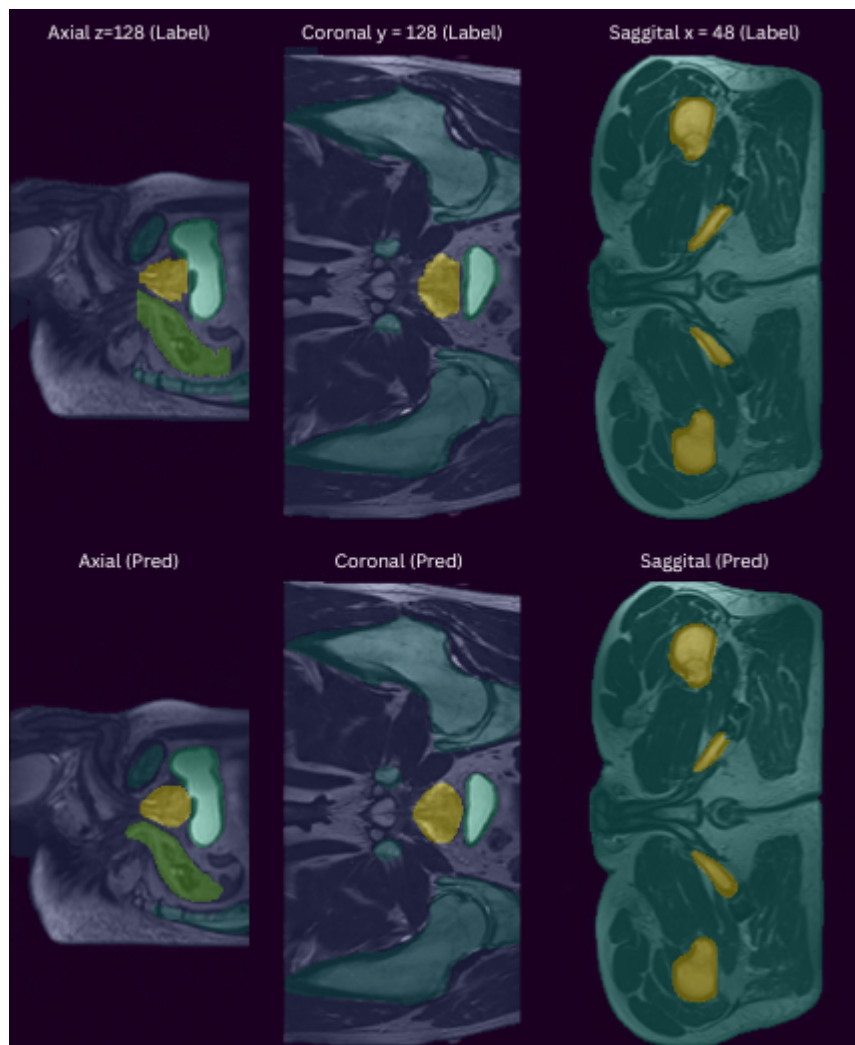
Training Results

The Improved UNet3D on the held-out test set, attained Dice scores of `[0.9983, 0.9880, 0.9221, 0.9458, 0.8591, 0.8491]`, all exceeding the 0.7 performance target. The model demonstrated strong generalization, accurately segmenting both major and smaller prostate structures, aided by attention, residual, SE, and ASPP modules.

Samples

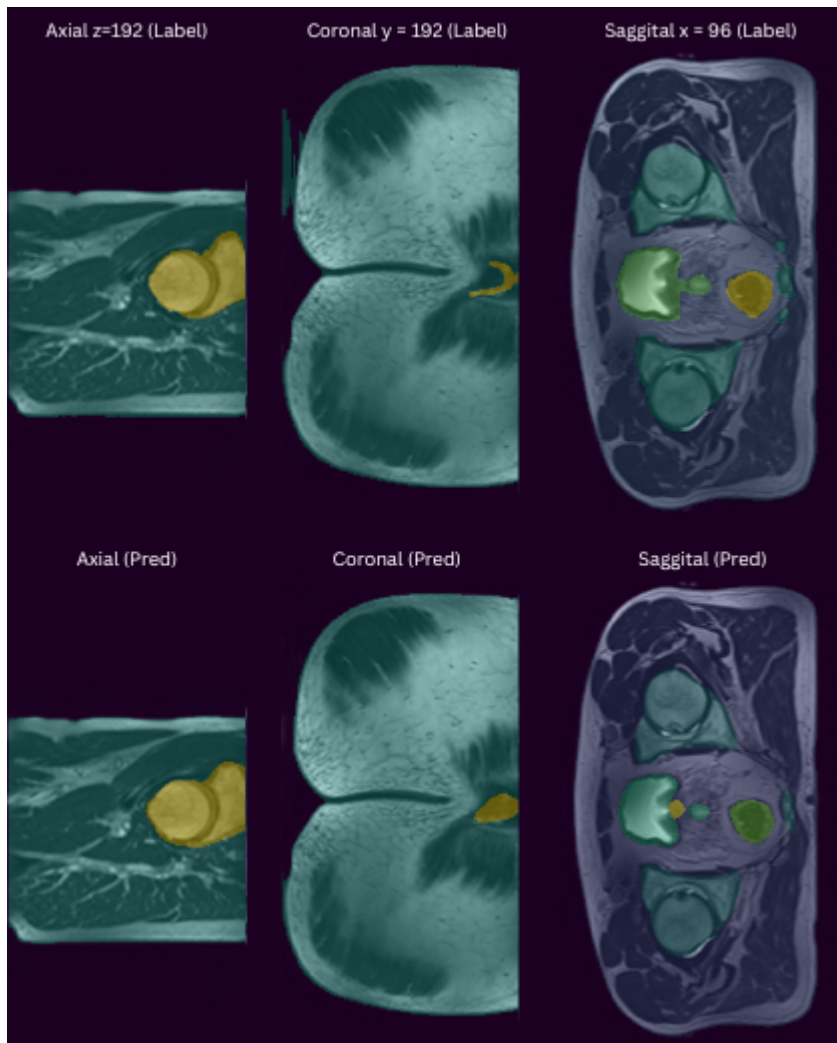
- The following figures illustrate qualitative segmentation results from the Improved UNet3D model on the Prostate 3D dataset.
- Each example shows the ground truth label (top row) and the predicted segmentation (bottom row) across three orthogonal MRI planes: axial, coronal, and sagittal.

Sample 1



- Axial: 128, Coronal: 128, Sagittal: 48
- Observation: Overall segmentation is good, with clear boundaries and consistent label continuity across the three planes.

Sample 2



- Axial: 192, Coronal: 192, Sagittal: 96
- Observation: Overall segmentation is strong. There are a few minor label mismatches noted in the sagittal plane, but they do not materially affect the overall quality.

Run Instructions

Set dataset paths In train.py, edit:

```
IMG_DIR = ROOT/semantic_MRs  
LAB_DIR = ROOT/semantic_labels_only
```

(Optional) Adjust hyperparameters

Dependencies

- Python 3.7+
- PyTorch 1.10+
- CUDA (for GPU support, optional)
- torchvision
- numpy
- tqdm

- matplotlib
- nibabel

References

- O. Cicek, A. Abdulkadir, S. S. Lienkamp, T. Brox, and O. Ronneberger, "3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation," in Medical Image Computing and Computer-Assisted Intervention – MICCAI 2016, ser. Lecture Notes in Computer Science, S. Ourselin, L. Joskowicz, M. R. Sabuncu, G. Unal, and W. Wells, Eds. Cham: Springer International Publishing, 2016, pp. 424–432.